Mikhail Razakov

Compiler / LLVM Engineer

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I work on compilers and static analysis. At MCST I implemented LLVM optimization passes and interprocedural IR analysis; I now work on SharpChecker at ISP RAS. My independent projects cover CFG/data-flow analysis, code generation, and compiler infrastructure.

EXPERIENCE

2026 — present

ISP RAS — Static Analysis Engineer

- Contributing to SharpChecker, ISP RAS's static-analysis platform for C#/.NET.

July — August 2026

MCST — Compiler Engineering Intern

- Implemented an LLVM LICM pass with loop analysis, side-effect/ speculative-safety checks, and hoisting of loop-invariant instructions to preheaders.
- Developed interprocedural global-variable analysis and legality checks for safe LLVM IR transformations.

PROJECTS

LLVM Interprocedural Global IV Optimization

An LLVM 22 pass for interprocedural analysis of linear global-variable evolution and safe IR transformation, using APInt, loop/dominator analysis, transitive effects, and must-execute conditions.

29 positive and negative regression tests, LLVM Verifier checks, idempotence checks, and before/after behavioral comparison.

DerefAfterNullAnalyzer

Roslyn ControlFlowGraph with forward fixed-point data-flow: Unknown/MaybeNull/NotNull states propagate across basic blocks and merge at join points.

Successors are reprocessed until the state stabilizes, producing diagnostics for potentially unsafe dereference paths.

UniversalToolchain

A modular .NET framework for language and compiler pipelines: parsing → intermediate representations → optimization passes → backend. Components use explicit contracts, with ordering and compatibility resolved before execution.

Language composition is separated from execution, with architectural contracts backed by executable verification and tests.

x86-64 Codegen & Register Allocation Lab

SSA-like IR, CFG/SSA validation, liveness/interference, register allocation with an independent assignment verifier, phi lowering, and SysV x86-64 codegen.

A reference interpreter and differential execution check generated-code semantics; the allocator contract is verified independently.

SKILLS

Compilers
C++, LLVM, LLVM IR, optimization passes, interprocedural analysis, CFG, SSA, dominators, loop analysis, LICM, legality analysis

Static / Program Analysis
C#, .NET, Roslyn, ControlFlowGraph, fixed-point data-flow analysis, state propagation and joins

Codegen / Tooling
register allocation, liveness analysis, x86-64, CMake, Linux, GitHub Actions, differential testing

EDUCATION

HSE University — Software Engineering, 2026–2030

RECOGNITION

LangDev'26 — Extensible Programming on .NET
Accepted for an oral presentation at LangDev'26.

UniversalToolchain — Baltic Science and Engineering Competition
First Degree Diploma and Grand Prize “Perfection as Hope” for UniversalToolchain.

MEPhI Junior
Absolute winner in 2025 and 2026; in 2026, First Degree Diploma, 96/100 for UniversalToolchain.

HSE Vysshaya Proba — Competitive & Industrial Programming
Prize-winner in HSE Vysshaya Proba in competitive and industrial programming.

Moscow / remote